WEEK – 9

1# A database is to be designed for sales information system. A product can be described with a unique product number, product name, selling price, manufacturer name. The product can sale to a particular client and each client have its own unique client ID, client name, client addresses, city, pin code and total balance to be required to paid. Each client orders to buy product from the salesman. In the order, it has unique sales order number, sales order date, client number and salesman number. Each salesman has the salesman number (unique), name, addresses, city, pin code, salary of the sales man.

a) Create ER Diagram and Schema diagram for the above system.
   * attached at end

b) Create the above tables and mention primary keys and the foreign keys

```sql
CREATE TABLE PRODUCT (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50),
    SellingPrice DECIMAL(10,2),
    ManufacturerName VARCHAR(50)
);

CREATE TABLE CLIENT (
    ClientID INT PRIMARY KEY,
    ClientName VARCHAR(50),
    Address VARCHAR(100),
    City VARCHAR(50),
    PinCode VARCHAR(10),
    TotalBalance DECIMAL(10,2)
);
```

```sql
CREATE TABLE SALES_ORDER (
    OrderID INT PRIMARY KEY,
    OrderDate DATE,
    ClientID INT,
    SalesmanID INT,
    FOREIGN KEY (ClientID) REFERENCES CLIENT(ClientID),
    FOREIGN KEY (SalesmanID) REFERENCES SALESMAN(SalesmanID)
);
```

```sql
CREATE TABLE ORDER_DETAILS (
    OrderID INT,
    ProductID INT,
    Quantity INT,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID) REFERENCES SALES_ORDER(OrderID),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID)
);
```

Results | Explain | Describe | Saved SQL | History

```
Table created.
```

0.09 seconds

```sql
CREATE TABLE SALESMAN (
    SalesmanID INT PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    City VARCHAR(50),
    PinCode VARCHAR(10),
    Salary DECIMAL(10,2)
);
```

c) Insert at least five relevant records in each of the created relations.

```sql
INSERT INTO PRODUCT VALUES
(1, 'Laptop', 45000, 'Dell'),
(2, 'Mobile', 15000, 'Samsung'),
(3, 'Tablet', 12000, 'Lenovo'),
(4, 'Printer', 8000, 'HP'),
(5, 'Mouse', 500, 'Logitech');

INSERT INTO CLIENT VALUES
(101, 'Amit', 'Street 1', 'ALIGARH', '202001', 2000),
(102, 'Ravi', 'Street 2', 'DELHI', '110001', 5000),
(103, 'Sahil', 'Street 3', 'MUMBAI', '400001', 3000),
(104, 'Anil', 'Street 4', 'ALIGARH', '202002', 1000),
(105, 'Kiran', 'Street 5', 'KANPUR', '208001', 4000);
```

Results | Explain | Describe | Saved SQL | History

```
5 row(s) inserted.
```

0.05 seconds

```sql
INSERT INTO SALESMAN VALUES
(201, 'Raj', 'Addr1', 'DELHI', '110001', 25000),
(202, 'Vikas', 'Addr2', 'ALIGARH', '202001', 20000),
(203, 'Neeraj', 'Addr3', 'MUMBAI', '400001', 30000),
(204, 'Aman', 'Addr4', 'KANPUR', '208001', 22000),
(205, 'Suresh', 'Addr5', 'DELHI', '110002', 27000);

INSERT INTO SALES_ORDER VALUES
(301, TO_DATE('2025-01-01', 'YYYY-MM-DD'), 101, 201),
(302, TO_DATE('2025-01-05', 'YYYY-MM-DD'), 102, 202),
(303, TO_DATE('2025-01-10', 'YYYY-MM-DD'), 103, 203),
(304, TO_DATE('2025-01-12', 'YYYY-MM-DD'), 104, 204),
(305, TO_DATE('2025-01-11', 'YYYY-MM-DD'), 105, 205);
```

```sql
INSERT INTO ORDER_DETAILS VALUES
(301, 1, 2),
(302, 2, 1),
(303, 3, 3),
(304, 4, 1),
(305, 5, 5);
```

d) d) Write following SQL queries based on created Database:

i) Retrieve names and cities of all the clients.

```
SELECT ClientName, City FROM CLIENT;
```

Results | Explain | Describe | Saved SQL | History

CLIENTNAME	CITY
Amit	ALIGARH
Ravi	DELHI
Sahil	MUMBAI
Anil	ALIGARH
Kiran	KANPUR

5 rows returned in 0.15 seconds Download

ii) Retrieve the available products.

```
SELECT * FROM PRODUCT;
```

Results | Explain | Describe | Saved SQL | History

PRODUCTID	PRODUCTNAME	SELLINGPRICE	MANUFACTURERNAME
1	Laptop	45000	Dell
2	Mobile	15000	Samsung
3	Tablet	12000	Lenovo
4	Printer	8000	HP
5	Mouse	500	Logitech

5 rows returned in 0.19 seconds Download

iii) Find the names of all clients having "a" as the second letter in their names.

```
SELECT ClientName
FROM CLIENT
WHERE ClientName LIKE '_a%';
```

Results | Explain | Describe | Saved SQL | History

Ravi
Sahil

2 rows returned in 0.02 seconds Download

iv) List of all the clients who are located in ALIGARH.

```
SELECT * FROM CLIENT
WHERE City = 'ALIGARH';
```

Results | Explain | Describe | Saved SQL | History

CLIENTID	CLIENTNAME	ADDRESS	CITY	PINCODE	TOTALBALANCE
101	Amit	Street 1	ALIGARH	202001	2000
104	Anil	Street 4	ALIGARH	202002	1000

2 rows returned in 0.02 seconds Download

v) Find the products whose selling price is less than or equal to 5000.

```
SELECT * FROM PRODUCT
WHERE SellingPrice <= 5000;
```

Results | Explain | Describe | Saved SQL | History

PRODUCTID	PRODUCTNAME	SELLINGPRICE	MANUFACTURERNAME
5	Mouse	500	Logitech

1 rows returned in 0.02 seconds Download

vi) List the products in sorted order of their selling price

```
SELECT * FROM PRODUCT
ORDER BY SellingPrice;
```

Results | Explain | Describe | Saved SQL | History

PRODUCTID	PRODUCTNAME	SELLINGPRICE	MANUFACTURERNAME
5	Mouse	500	Logitech
4	Printer	8000	HP
3	Tablet	12000	Lenovo
2	Mobile	15000	Samsung
1	Laptop	45000	Dell

5 rows returned in 0.01 seconds Download